ELEC-PE-06 Ericsson PE Node Hardware/Environmental Fault Troubleshooting SOP

Node: ELEC-PE-06 | **Vendor:** Ericsson | **Severity:** Major | **Alarm Code:** ALARM_3291

Description

The ELEC-PE-06 node, situated in the Electronic City Hub, is an Ericsson PE node responsible for providing critical connectivity services. The failure domain of hardware/environmental faults, as indicated by ASIC memory parity errors, line card crashes, and power supply redundancy loss, is a significant concern. This SOP is designed to guide the troubleshooting and resolution of the ALARM_3291 major alarm, which is triggered by these faults. The underlying protocol or hardware mechanics responsible for this failure domain can be attributed to the complex interplay between the node's hardware components, including the Application-Specific Integrated Circuit (ASIC), line cards, and power supplies. The Ericsson PE node's architecture relies heavily on the reliable operation of these components to ensure seamless data transmission. However, in the event of hardware or environmental failures, the node's ability to maintain connectivity is compromised, leading to the ALARM_3291 alarm. This SOP outlines the necessary diagnostic, resolution, and verification steps to identify and rectify the root cause of the failure.

Pre-requisite Checks

- pre-requisite: Verify the Electronic City Hub's environmental conditions, including temperature, humidity, and power supply status.

- pre-requisite: Check the redundant link status to ensure that the node is not experiencing any link failures.

- pre-requisite: Dispatch remote hands to the Electronic City Hub to assist with physical checks and repairs, if necessary.

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show asp switchover state, Expected Output: SWITCHOVER STATE: Active, Description: This command is used to verify the node's switchover state, which helps identify if the node has experienced a switchover event due to a hardware or environmental failure.
```

```
Command: show asp asp state, Expected Output: ASIC STATE: UP,
Description: This command is used to verify the ASIC's operational
state, which helps identify if the ASIC is experiencing any errors or
issues.

Command: show asp alarm, Expected Output: ALARM STATE: Active,
Description: This command is used to verify the node's alarm state,
which helps identify if the node is experiencing any alarms or errors.
```

Resolution Steps

Step 1: Step: Replace the faulty ASIC, Command: replace asp asp, Description: This step involves replacing the faulty ASIC with a new one to rectify the hardware failure. The 'replace asp asp' command is used to initiate the replacement process.

Step 2: Step: Replace the faulty line card, Command: replace linecard , Description: This step involves replacing the faulty line card with a new one to rectify the hardware failure. The 'replace linecard ' command is used to initiate the replacement process, where is the slot number of the faulty line card.

Step 3: Step: Verify power supply redundancy, Command: show power supply, Description: This step involves verifying the power supply redundancy to ensure that the node has sufficient power supply redundancy to maintain connectivity.

Post-Check Verification

```
Command: show asp switchover state, Expected Output: SWITCHOVER STATE:
Standby, Description: This command is used to verify that the node has
successfully switched over to the standby state after the resolution
steps.

Command: show asp asp state, Expected Output: ASIC STATE: UP,
Description: This command is used to verify that the ASIC is
operational after the resolution steps.

Command: show asp alarm, Expected Output: ALARM STATE: Clear,
Description: This command is used to verify that the node's alarm
state has cleared after the resolution steps.
```

Roll-back Procedures

- Step: Undo the replacement of the faulty ASIC, Command: undo replace asp asp, Description: This step involves undoing the replacement of the faulty ASIC to roll back to the previous state. The 'undo replace asp asp' command is used to initiate the undo process.

- Step: Undo the replacement of the faulty line card, Command: undo replace linecard , Description: This step involves undoing the replacement of the faulty line card to roll back to the previous state. The 'undo replace linecard ' command is used to initiate the undo process, where is the slot number of the faulty line card.